

Probability $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$

$$1) \mathbb{P}(\Omega) = 1$$

$$2) \sum_{i=1}^{\infty} \mathbb{P}(A_i) \quad A_i \cap A_j = \emptyset \quad i \neq j$$

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i) \quad \text{for all events?}$$

or assuming independent?

Conditional Probability

Restrict total sample space

from Ω to B

$$\mathbb{P}(B) > 0,$$

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

$$\mathbb{P}(A|B)\mathbb{P}(B) = \mathbb{P}(A \cap B)$$

$$\mathbb{P}(A|B^c)\mathbb{P}(B^c) = \mathbb{P}(A \cap B^c)$$

$$\mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c) = \mathbb{P}(A)$$

$$\mathbb{P}(A|B)\mathbb{P}(B) + \mathbb{P}(A|B^c)\mathbb{P}(B^c) = \mathbb{P}(A)$$

Law of total probability

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A|B_i)\mathbb{P}(B_i)$$

if $\sum B_i$ pairwise disjoint

$$\text{and } \bigcup_{i=1}^n B_i = \Omega$$

for infinite prob space as $n \rightarrow \infty$ s.t. $\bigcup_{i=1}^{\infty} B_i = \Omega$

Bayes Rule

$$\mathbb{P}(B_i|A) = \frac{\mathbb{P}(A|B_i)\mathbb{P}(B_i)}{\sum_{j=1}^n \mathbb{P}(A|B_j)\mathbb{P}(B_j)}$$

1. Probability Measure and Axioms

A probability space is a measure space $(\Omega, \mathcal{F}, \mathbb{P})$ with total mass 1:

- Ω is the sample space.
- $\mathcal{F} \subseteq 2^\Omega$ is a σ -algebra of measurable events.
- $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is the probability measure satisfying the Kolmogorov axioms:

1. **Normalization:**

$$\mathbb{P}(\Omega) = 1$$

2. **Countable Additivity (σ -additivity):**

For any countable collection of pairwise disjoint events $\{A_i\}_{i=1}^{\infty} \subset \mathcal{F}$ where $A_i \cap A_j = \emptyset$ for all $i \neq j$:

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$$

Sidebar Query: For all events? Or assuming independence?

Clarification: This holds for **pairwise disjoint (mutually exclusive) events**, not independent events. Mutual exclusivity ($A_i \cap A_j = \emptyset$) means the events cannot occur simultaneously. Independence ($\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$) is an orthogonal concept; if two non-trivial events are mutually exclusive ($\mathbb{P}(A \cap B) = 0$), they are fundamentally **dependent** because knowing one occurred guarantees the other did not.

2. Conditional Probability

Conditioning on an event $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$ restricts the effective sample space from Ω to B , rescaling the measure so that $\mathbb{P}(B|B) = 1$:

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

$\Downarrow$

$$\mathbb{P}(A \cap B) = \mathbb{P}(A|B)\mathbb{P}(B)$$

Similarly, for the complement $B^c = \Omega \setminus B$:

$$\mathbb{P}(A \cap B^c) = \mathbb{P}(A|B^c)\mathbb{P}(B^c)$$

Since $A = (A \cap B) \cup (A \cap B^c)$ and $(A \cap B) \cap (A \cap B^c) = \emptyset$, additivity implies:

$$\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c)$$

$$\mathbb{P}(A) = \mathbb{P}(A|B)\mathbb{P}(B) + \mathbb{P}(A|B^c)\mathbb{P}(B^c)$$

3. Law of Total Probability

Let $\{B_i\}_{i=1}^n$ form a measurable partition of Ω , meaning $B_i \cap B_j = \emptyset$ for $i \neq j$, $\bigcup_{i=1}^n B_i = \Omega$, and $\mathbb{P}(B_i) > 0$ for all i . Then:

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A|B_i)\mathbb{P}(B_i)$$

4. Bayes' Rule

For an arbitrary partition element B_k and an event A with $\mathbb{P}(A) > 0$:

$$\mathbb{P}(B_k|A) = \frac{\mathbb{P}(A \cap B_k)}{\mathbb{P}(A)}$$

$$\mathbb{P}(B_k|A) = \frac{\mathbb{P}(A|B_k)\mathbb{P}(B_k)}{\sum_{i=1}^n \mathbb{P}(A|B_i)\mathbb{P}(B_i)}$$

Clarification: For infinite prob space as $n \rightarrow \infty$, switch to $\int$ rather than $\sum$?

Clarification:

- Countably Infinite Partitions:** If the partition is countably infinite ($\{B_i\}_{i=1}^{\infty}$ with $\bigcup_{i=1}^{\infty} B_i = \Omega$), the discrete summation still applies directly via monotone convergence / σ -additivity:

$$\mathbb{P}(A) = \sum_{i=1}^{\infty} \mathbb{P}(A|B_i)\mathbb{P}(B_i)$$

- Continuous State Spaces (Uncountable):** When conditioning on a continuous random variable Θ with probability density function $f_\Theta(\theta)$, discrete conditioning $\mathbb{P}(\cdot|\Theta = \theta)$ fails because $\mathbb{P}(\Theta = \theta) = 0$. In this regime, the summation transitions to an integral using Radon-Nikodym derivatives / regular conditional probabilities:

$$\mathbb{P}(A) = \int_{\mathbb{R}} \mathbb{P}(A|\Theta = \theta) f_\Theta(\theta) d\theta$$

and the continuous Bayes' formula for the posterior density $f_{\Theta|X}(\theta|x)$ becomes:

$$f_{\Theta|X}(\theta|x) = \frac{f_{X|\Theta}(x|\theta)f_\Theta(\theta)}{\int_{\mathbb{R}} f_{X|\Theta}(x|u)f_\Theta(u) du}$$

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Bayes Rule

$$P(B_i | A) = \frac{P(A \cap B_i)}{P(A)}$$

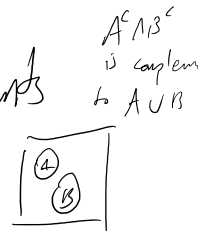
$$P(B_i | A) = \frac{P(A | B_i) P(B_i)}{\sum_{i=1} P(A | B_i) P(B_i)}$$

prob space
as $n \rightarrow \infty$
converge to
integral then $\int$

Independence

if A & B independent events

$$P(A \cap B) = P(A)P(B)$$



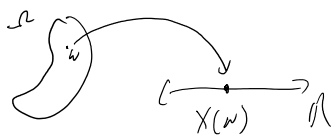
$$\frac{P(A \cap B)}{P(B)} = P(A) = P(A | B)$$

B gives no new info on A

Random Variables

measurable functions $\Omega \rightarrow \mathbb{R}$

takes event A & assigns a value



Like
counting
number
of heads
in coin tosses?
or combined
value of dice?

CDF

$$F_X(x) = P(X \leq x)$$

non-decreasing
right continuous

$$F_X(x) \rightarrow 0 \text{ as } x \rightarrow -\infty$$

$$F_X(x) \rightarrow 1 \text{ as } x \rightarrow \infty$$

Discrete Random Variable

for x taking values in $\{x_i\}_{i=1}^N$

$$F_X(x) = P(X \leq x) = \sum_{x_i \leq x} P(X = x_i)$$

Continuous Random Variable

1. Independence of Events

Two events $A, B \in \mathcal{F}$ are statistically independent if and only if:

$$P(A \cap B) = P(A)P(B)$$

Assuming $P(B) > 0$, this definition is equivalent to:

$$\frac{P(A \cap B)}{P(B)} = P(A) \Leftrightarrow P(A | B) = P(A)$$

Conditioning on B yields no informational gain regarding the realization of A .

Sidebar Annotation Note: " $A^c \cap B^c$ is complement to $A \cup B$ " and the Venn diagram with disjoint circles.

Clarification: By De Morgan's Laws, $(A \cup B)^c = A^c \cap B^c$ is true. However, the accompanying sketch of two non-overlapping circles depicts **disjoint/mutually exclusive** sets ($A \cap B = \emptyset$), not independent sets. As established, if $P(A) > 0$ and $P(B) > 0$, disjoint events can **never** be independent because $P(A \cap B) = 0 \neq P(A)P(B)$.

2. Random Variables

Formally, a random variable X defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a **measurable function**:

$$X: \Omega \rightarrow \mathbb{R}$$

such that the preimage of any Borel set $B \in \mathcal{B}(\mathbb{R})$ is $\mathcal{F}$ -measurable:

$$X^{-1}(B) = \{\omega \in \Omega: X(\omega) \in B\} \in \mathcal{F}$$

Sidebar Query: Like counting number of heads in coin tosses? Or combined value of dice?

Clarification: Yes, precisely:

- For n coin tosses, the sample space is $\Omega = \{H, T\}^n$, and $X(\omega) = \sum_{i=1}^n \mathbf{1}_{\{\omega_i = H\}}$ maps each sequence ω to the integer count of heads in $\{0, 1, \dots, n\}$.
- For rolling two dice, $\Omega = \{(i, j): 1 \leq i, j \leq 6\}$, and $X(i, j) = i + j$ maps each pair to an integer sum in $\{2, \dots, 12\}$.

3. Cumulative Distribution Function (CDF)

The Cumulative Distribution Function $F_X: \mathbb{R} \rightarrow [0, 1]$ completely characterizes the induced probability measure of X :

$$F_X(x) = P(X \leq x) = \mathbb{P}(\{\omega \in \Omega: X(\omega) \leq x\})$$

Fundamental Properties:

- Monotonicity:** $F_X(x_1) \leq F_X(x_2)$ whenever $x_1 \leq x_2$.
- Right-continuity:** $\lim_{h \rightarrow 0^+} F_X(x + h) = F_X(x)$ for all $x \in \mathbb{R}$.
- Asymptotic Limits:**

$$\lim_{x \rightarrow -\infty} F_X(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow \infty} F_X(x) = 1$$

4. Discrete and Continuous Random Variables

Discrete Random Variables:

If the support of X is a countable or finite set $\{x_i\}_{i=1}^N \subset \mathbb{R}$, the CDF is a step function:

$$F_X(x) = P(X \leq x) = \sum_{x_i \leq x} P(X = x_i)$$

Continuous Random Variables:

If $F_X(x)$ is absolutely continuous with respect to the Lebesgue measure, there exists a non-negative, integrable probability density function (PDF) $f_X: \mathbb{R} \rightarrow [0, \infty)$ such that:

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(t) dt$$

By the Fundamental Theorem of Calculus (almost everywhere):

Continuous Random Variable
 $F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^x f_X(t) dt$

$$F'_X(x) = f_X(x)$$

Joint CDF $F_{X,Y}$

$$F_{X,Y}(x,y) = \mathbb{P}(X \leq x, Y \leq y)$$

$$\text{If } F_{X,Y}(x,y) = \int_{-\infty, x] \times [-\infty, y]} f_{X,Y}(x,y) dx dy$$

then marginal density

$$f_X(x) = \int f_{X,Y} dy$$

$$f_Y(y) = \int f_{X,Y} dx$$

Independence of Random Variables

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A) \mathbb{P}(Y \in B)$$

Expectation of Random Variable

Discrete

$$\mathbb{E}[X] = \sum_x x \mathbb{P}[X=x] = \sum_x x p_X(x)$$

Continuous

$$\mathbb{E}[X] = \int x f_X(x) dx$$

$$\mathbb{E}[g(x)] = \sum g(x) p_X(x)$$

$$\text{or } = \int g(x) f_X(x) dx$$

Variance is special case of $g(x)$

$$\mathbb{E}[(X - \mathbb{E}[X])^2] \quad \text{Second centered moment known}$$

$$F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^x f_X(t) dt$$

By the Fundamental Theorem of Calculus (almost everywhere):

$$f_X(x) = F'_X(x) = \frac{d}{dx} F_X(x)$$

5. Joint Distributions and Marginal Densities

For a random vector $(X, Y) \in \mathbb{R}^2$, the **Joint CDF** is:

$$F_{X,Y}(x,y) = \mathbb{P}(X \leq x, Y \leq y)$$

If (X, Y) is jointly continuous with joint density $f_{X,Y}(x, y)$:

$$F_{X,Y}(x,y) = \iint_{(-\infty, x] \times (-\infty, y]} f_{X,Y}(u,v) du dv = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(u,v) dv du$$

Marginal Densities:

Integrating out the nuisance variable gives the marginal densities:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx$$

6. Independence of Random Variables

Two random variables X and Y are independent if and only if their generated σ -algebras $\sigma(X)$ and $\sigma(Y)$ are independent:

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A) \mathbb{P}(Y \in B) \quad \forall A, B \in \mathcal{B}(\mathbb{R})$$

Equivalently, in terms of CDFs and PDFs:

$$F_{X,Y}(x,y) = F_X(x) F_Y(y) \quad \forall x, y \in \mathbb{R}$$

$$f_{X,Y}(x,y) = f_X(x) f_Y(y) \quad \text{a.e.}$$

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1. Expectation of a Random Variable

The mathematical expectation (first raw moment) represents the probability-weighted average of all possible realizations of a random variable X :

• Discrete Case:

$$\mathbb{E}[X] = \sum_x x \mathbb{P}(X=x) = \sum_x x p_X(x)$$

• Continuous Case:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

Law of the Unconscious Statistician (LOTUS):

For any measurable function $g: \mathbb{R} \rightarrow \mathbb{R}$:

$$\mathbb{E}[g(X)] = \sum_x g(x) p_X(x) \quad \text{or} \quad \mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

2. Variance and Covariance

Variance:

Variance measures the expected squared deviation from the mean, defined as the special case $g(X) = (X - \mathbb{E}[X])^2$:

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Sidebar Annotation: Second centered moment from MGF.

Deeper Context & Derivation: The Moment-Generating Function (MGF) is defined as $M_X(t) = \mathbb{E}[e^{tX}]$.

$$\mathbb{E}[(X - \mathbb{E}[X])^2] \quad \text{Second centered moment from MGF}$$

$$= \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Covariance

$$\mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

If X & Y independent

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$$

if not independent

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

Conditional Expectation

$$\mathbb{E}[X | Y=y] = \sum_x x \mathbb{P}[X=x | Y=y]$$

$$= \int x f_{X|Y}(x=y) dx$$

by averaging at x , cond exp

$\mathbb{E}[X | Y=y]$ is function of y being the realization of Y

once we take conditional, variability is driven by y

$\mathbb{E}[X|Y]$ is random variable by law of iterated expectations,

$$\mathbb{E}[\mathbb{E}[X|Y]] = \mathbb{E}[X]$$

Tower property

Sidebar Annotation: Second centered moment from MGF.

Deeper Context & Derivation: The Moment-Generating Function (MGF) is defined as $M_X(t) = \mathbb{E}[e^{tX}]$.

- The raw moments are obtained by differentiating evaluated at $t = 0$:

$$\mathbb{E}[X^k] = \left. \frac{d^k M_X(t)}{dt^k} \right|_{t=0}$$

- The variance is the **second central moment** $\mu_2 = \mathbb{E}[(X - \mu)^2]$. It can be extracted directly via the **Cumulant Generating Function** $K_X(t) = \ln M_X(t)$:

$$\text{Var}(X) = K_X''(0) = \left. \frac{d^2 \ln M_X(t)}{dt^2} \right|_{t=0}$$

$$= \frac{M_X''(0)M_X(0) - [M_X'(0)]^2}{[M_X(0)]^2} = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Covariance:

Covariance measures the joint linear association between two random variables:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

Variance of Sums:

- General Case:**

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

- If X and Y are **independent** (or simply uncorrelated, $\text{Cov}(X, Y) = 0$):

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

3. Conditional Expectation

For a given realization $Y = y$, the conditional expectation $\mathbb{E}[X | Y = y]$ is a **deterministic scalar constant**:

- Discrete Case:**

$$\mathbb{E}[X | Y = y] = \sum_x x \mathbb{P}(X = x | Y = y)$$

- Continuous Case:**

$$\mathbb{E}[X | Y = y] = \int_{-\infty}^{\infty} x f_{X|Y}(x | y) dx$$

$$\text{where } f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

By integrating (or summing) out x , $\mathbb{E}[X | Y = y]$ becomes a deterministic function of the fixed value y , written as $h(y)$.

4. Conditional Expectation as a Random Variable & Tower Property

When conditioning on the random variable Y rather than a specific realized value y , the object:

$$\mathbb{E}[X | Y] = h(Y)$$

is itself a **random variable** measurable with respect to the σ -algebra $\sigma(Y)$. Its randomness is completely driven by Y .

Law of Iterated Expectations (Tower Property):

$$\mathbb{E}[\mathbb{E}[X | Y]] = \mathbb{E}[X]$$

Deeper Context & Measure-Theoretic View:

- Underlying Mechanism:** The tower property is the expectation-level equivalent of the Law of Total Probability:

$$\mathbb{E}[\mathbb{E}[X | Y]] = \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} x f_{X|Y}(x | y) dx \right) f_Y(y) dy$$

$$= \int_{-\infty}^{\infty} x \left(\int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \right) dx = \int_{-\infty}^{\infty} x f_X(x) dx = \mathbb{E}[X]$$

- General Conditioning:** For any sub- σ -algebra $\mathcal{G} \subseteq \mathcal{F}$,

$$\mathbb{E}[\mathbb{E}[X|Y]] = \mathbb{E}[X]$$

Tower property

uses law of total probability

$$= \int_{-\infty}^{\infty} x \left(\int_{-\infty}^{\infty} f_{X,Y}(x,y) dy \right) dx = \int_{-\infty}^{\infty} x f_X(x) dx = \mathbb{E}[X]$$

- 2. General Conditioning:** For any sub- σ -algebra $\mathcal{G} \subseteq \mathcal{F}$, $\mathbb{E}[X | \mathcal{G}]$ is defined as the unique $\mathcal{G}$ -measurable random variable satisfying:

$$\int_G \mathbb{E}[X | \mathcal{G}] d\mathbb{P} = \int_G X d\mathbb{P} \quad \forall G \in \mathcal{G}$$

Setting $G = \Omega$ yields $\mathbb{E}[\mathbb{E}[X | \mathcal{G}]] = \mathbb{E}[X]$.

- 3. Tower Hierarchy:** If $\mathcal{H} \subseteq \mathcal{G} \subseteq \mathcal{F}$, coarser information dominates:

$$\mathbb{E}[\mathbb{E}[X | \mathcal{G}] | \mathcal{H}] = \mathbb{E}[X | \mathcal{H}]$$

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